Week 11: 09– 13 November 2026

Subject: Mathematics **Unit Title:** Linear Functions and Equations **Grade :** 8 Pre AP Algebra 1

Lessons الحصص	Learning targets أهداف التعلم	Assessment tools and strategies التقييم Formative/تكويني Summative /ختامي Differentiation التمايز	Resources and links for Parents المصادر و الروابط المخصصة لأولياء الأمور
Lesson 1: 1.7 Sequences, Functions, and Graphs **Lesson 2 1.7 Sequences, Functions, and Graphs** **Lesson 3: 1.8 Interpreting Different Forms of a Linear Equation** **Lesson 4 1.8 Interpreting Different Forms of a Linear Equation**	• 1.1.2 Construct and use arithmetic sequences to represent linear relationships. • 1.1.5 Construct and apply linear functions to solve problems given mathematical or applied contexts. • 1.2.1 Identify and interpret key characteristics of linear functions and their graphs. • 1.2.2 Generate an algebraic representation of a linear function given a verbal description, graphical representation, or tabular representation. • 1.2.3 Translate between different algebraic forms of a linear equation.	1.1 Flexible display of information (size, color, font, layout). 2.1 Clarify vocabulary and symbols using visuals, connections, and supports. 8.1 Make goals clear, visible, and supported by short-term objectives.	Reflection prompts, notebooks The resources will be available on student's Teams channel
Teacher's notes/ Reminders ملاحظات المعلم والتذكير	**Formatives take place during lesson time, ALWAYS be prepared.** Bring your stationary: Penic/Pen and Calculator and be on time. **Remember** we will be using our notebooks. ***Devices will only be used on Friday's, unless otherwise communicated by your teacher.*** Communication takes place via **TEAMS** or **Toddle**. We will be introducing our AP Platform soon, **College Board** **Summative date Boys School: Tuesday, November 10, 2026** **Homework** is assigned through TEAMS/TODDLE as practice from teachers.		

Technology Integration دمج التكنولوجيا	Resources shared on Microsoft Teams, College Board
Cross-curricular links الروابط بين المواد الدراسية	**Science**: Measurement and unit conversions. **Physics**: Speed, rate, and proportional relationships. **Geography**: Metric and customary measurement systems. **Engineering**/Design: Using mathematical models to solve real-world problems.

External Assessment Integration دمج التقييمات الخارجية	All content is covered in AP Past Exams/ College Board Assessments
Wellbeing/ ATL Skills الصحة النفسية/ أساليب التعلم	**Wellbeing:** • Confidence in solving real-world maths problems. • Resilience when learning from mistakes. • Collaboration through shared investigations and discussions. • Self-awareness by setting and reflecting on learning goals. **IB ATL Skills:** • **Thinking:** Analyze patterns and relationships. • **Communication:** Explain and justify mathematical reasoning. • **Research:** Collect, organize, and interpret data. • **Social:** Collaborate and discuss problem-solving strategies
Standards المعايير	• **A.CED.2** • **F.IF.3, 5** • **F.BF.1, 2** • **F.LE.2**
Inquiry/ TOK Question البحث والاستقصاء أسئلة TOK	How can different representations of a linear relationship help us understand and solve real-world problems?
Success Criteria معايير النجاح (Measurable)	Lesson 1 **All Students Must:** identify arithmetic sequences and determine the common difference between consecutive terms. **Most Students Should:** represent arithmetic sequences using tables and identify patterns in the sequence. **Some Students Could:** write explicit rules for arithmetic sequences and explain how the rule relates to the pattern. Lesson 2: **All Students Must:** connect arithmetic sequences to linear functions and represent them graphically. **Most Students Should:** determine the rate of change and initial value from tables, graphs, and sequences. **Some Students Could:** construct and apply linear functions to solve real-world problems involving constant rates of change. Lesson 3: **All Students Must:** identify slope and y-intercepts from graphs, tables, and equations. **Most Students Should:** write linear equations in slope-intercept form and interpret the meaning of the slope and intercept in context. **Some Students Could:** generate linear equations from graphical, tabular, and verbal representations. Lesson 4:

	All Students Must: recognize and interpret linear equations written in different forms. **Most Students Should:** translate between slope-intercept form and point-slope form. **Some Students Could:** justify the equivalence of different forms of a linear equation and select the most appropriate form for solving a problem or modeling a situation.
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